CODE:

```
def simulate (total_frames,
    window_size, lost_frame):
        base = 0
        next_seq = 0.
        while base < total_frames:
            window_end = min (base + window_size - 1, total_frames-1)
        print ("Sender Window : [Frame ", base, "-Frame ", window_end, "]")
        while next_seq < base + window_size and next_seq <
                                                total_frames:
            print ("Node A -> Sending Frame ", next_seq)
            if next_seq == lost_frame:
                print ("          [Frame ", next_seq, "LOST in
                                                TRANSIT!)")
                has
            next_seq += 1
        if lost_frame is not None and base <= lost_frame < next_
                                                    seq:
            print ("Node B --> Frame, lost_frame, "missing.
                    Discarding out-of-order frames after it.")
            print ("Node A --> Time out! Resending window starting
                    from Frame", lost_frame)
            next_seq = lost_frame
            lost_frame = None.
            continue.
        for f in range (base, next_seq):
            print ("Node B --> Received Frame", f, "Sending ACK
                                                    f")
        print ("Node A --> ACK ", next_seq-1, "received. Sliding
                                        window forward.")
        base = next_seq.
```

Lab book Ex:07 FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

AIM:

To write a program to implement flow control at the Data link layer using the sliding window protocol, and to simulate the flow of frame from one node to another.

ALGORITHM:

1. Start the program.
2. Read the total no. of frames to be sent (N), the sender's window size (W), and (optionally) a frame no. to simulate as lost during transaction. transmission.
3. Initialize base = 0 (sequence number of the oldest unacknowledged frame) and nextSeq = 0 (sequence no. of the next frame to be sent).
4. Repeat while base < N:
   (a) Determine the current sender window as the range [base, base + w-1].
   (b) Send all frames with sequence numbers from nextSeq up to (base + W-1), incrementing nextSeq after each frame, since the window restricts the sender to at most W outstanding frames at any time.
   (c) If the lost frame lies within the sent frames just sent:
      (i) The reciever discards the lost frame and any subsequent out-of-order frames.
      (ii) The sender times out and retransmits all frames starting from the lost frame (Go-Back-N)
   (d) Else (no loss in this batch):
      (i) The receiver accepts each frame in order and sends a cummulative acknowledgement (ACK) for it.
      (ii) On receiving the ACK for the last frame sent, the sender slides its window forward: base = nextSeq.
5. Repeat step 4. until all N frames have been transmitted and acknowledged.
6. Stop the program.

```
print ("All ", total_frames, "frames successfully transmitted
       from Node A to Node B.")

def main():
    print ("SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION")
    total_frames = int(input("Enter total no. of frames to send: "))
    window_size = int(input("Enter sender's window size"))
    lost_frame = int(input(f"Enter frame no. to simulate as
                 Lost \n" f"(0 to {total_frame-1},
                 enter -1 for no loss) : "
    if lost_frame == -1:
        lost_frame = None.
    Simulate (total_frames, window_size, lost_frame)
if __name__ == "__main__":
    main().
```

OUTPUT:

```
SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION
Enter total no. of frames to send : 8
Enter sender's window size : 4.
Enter frame no. to simulate as LOST.
(0 to 7, enter -1 for no loss : 5

Sender Window: [Frame 0 - Frame 3]
Node A -> Sending Frame 0
Node A -> Sending Frame 1
Node A -> Sending Frame 2
Node A -> Sending Frame 3
Node B -> Received Frame 0, sending ACK 0
Node B -> Recived Frame 1, sending ACK 1
Node B -> Received Frame 2, sending ACK 2
Node B -> Received Frame 3, sending ACK 3.
Node A -> ACK 3 received, sliding window forward.
```

RESULT:

The program to implement flow control at the Data Link Layer using the Sliding Window Protocol was successfully written in Python, executed and verified by simulating the flow of frames between 2 nodes (Node A as sender, and Node B as receiver).